

where v_1 represents the starting speed of the ball in the x (horizontal) direction. Moreover, the height of a thrown ball (y), according to Galileo, can be calculated at time t as

$$y = v_2 t + \frac{at^2}{2}$$

where v_2 represents the starting speed of the ball in the y (vertical) direction, and a represents the constant downward acceleration due to gravity (which will be about -9.81 if we are working in metric units). When we substitute the first equation into the second equation, we find that the height of a thrown ball (y) relates to the horizontal position of the ball (x) as follows:

$$y = \frac{v_2}{v_1} x + \frac{ax^2}{2v_1^2}$$

We can use Galileo's equations to model a hypothetical ball's trajectory in Python using the function in [Listing 1-1](#). The specific polynomial in [Listing 1-1](#) is appropriate for a ball whose initial horizontal speed is about 0.99 meters per second, and whose initial vertical speed is about 9.9 meters per second. You can feel free to try other values for v_1 and v_2 to model any type of throw that interests you.

```
def ball_trajectory(x):  
    location = 10*x - 5*(x**2)  
    return(location)
```

[Listing 1-1](#): A function for calculating the trajectory of a ball

We can plot the function in [Listing 1-1](#) in Python to see what, approximately, a ball's trajectory should look like (ignoring air resistance and other negligible factors). We'll import some plotting capabilities from a module called `matplotlib` in the first line. The `matplotlib` module is one of many third-party modules we'll import in code throughout this book. Before you use a third-party module, you'll have to install it. You can install `matplotlib` and any other third-party modules by following the instructions at <http://automatetheboringstuff.com/2e/appendixa/>.